Identifying Fairness Concerns and Valuations of Products Using Pay-As-You-Want Pricing

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1. Introduction
2. Literature
3. Data
4. Research Design

1 Introduction

2 Literature

3 Data

4 Research Design

A motivating example

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- Classic models suggest consumers pay as little as possible
- However, evidence shows consumers sometimes pay more

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- Arts: the rock band Radiohead and Bandcamp
- Restaurants: Der Wiener Deewan
- Live streaming: YouTube Live and Twitch
- A broader context: paying tips

Research questions

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  - How to identify the distribution of consumers' willingness to pay more in the market?
  - How to identify the distribution of willingness to pay for consumers who are willing to pay more?

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Literature

Key contributors of the profitability of PAYW

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Methods used in the PAYW literature

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- Analytical models (Chen et al. 2017, Kim et al. 2022)
- Experiments (Gneezy et al. 2010; Schmidt et al. 2015)
- The first one to study PAYW in a large-scale, real world empirical context

The most related paper

Chen, Y., Koenigsberg, O., & Zhang, Z. J. (2017). Pay-as-you-wish pricing. *Marketing Science*, 36(5), 780-791.

The most related paper

- Consumer i's utility from purchasing a product is given by

$$u_i = r_i - p_i - \beta_i \max\{p_i - r_{i0}, 0\} - \gamma_i \max\{r_{i0} - p_i, 0\},$$

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  - r_i is consumption utility
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  - $\beta_i, \gamma_i \geq 0$ captures consumers' distility toward disadvantageous and advantageous inequality (Fehr and Schmidt 1999)

The most related paper

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Main findings:

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Main findings:

- PAYW can help a firm achieve maximal market penetration, implement price discrimination, and moderate price competition.

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Bandcamp data

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  - Recent promotion activities
- Information about users
  - Followers, followings, collections, and wish lists
  - Interested tags

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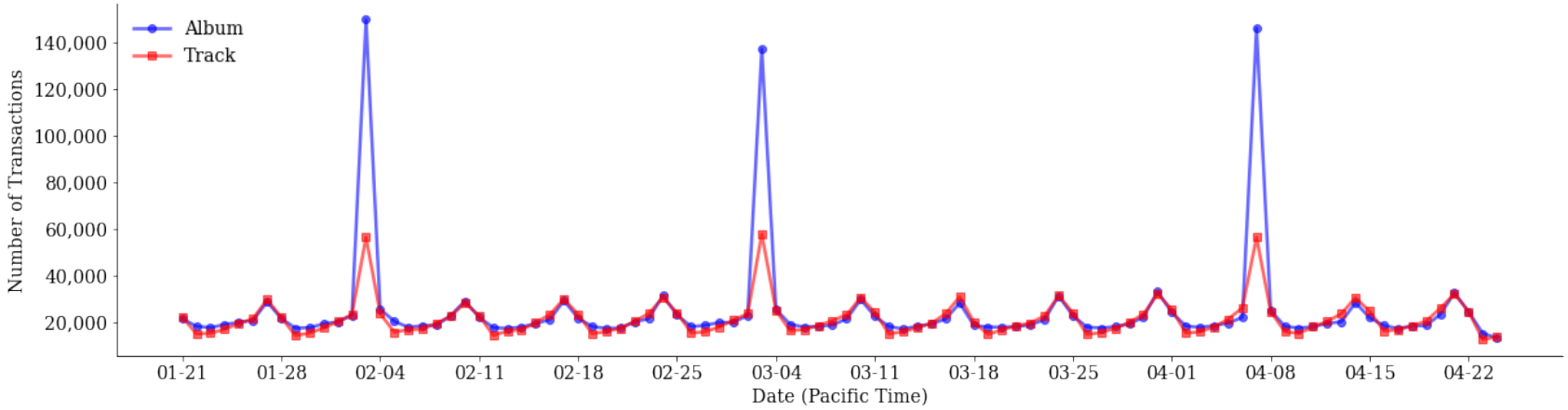

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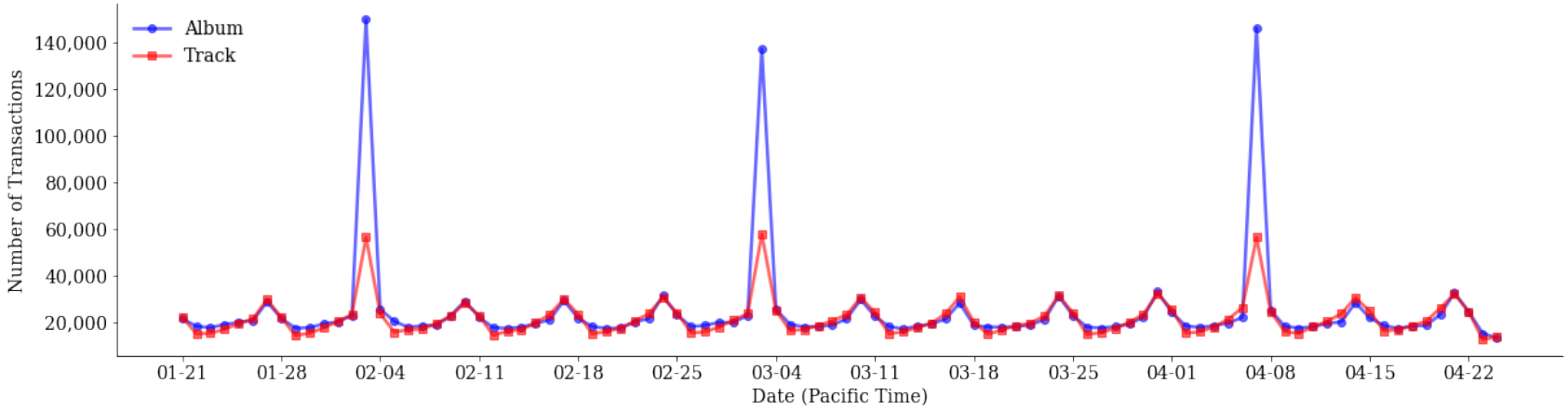

The descriptive analysis shows that price variations for most products are ignorable.

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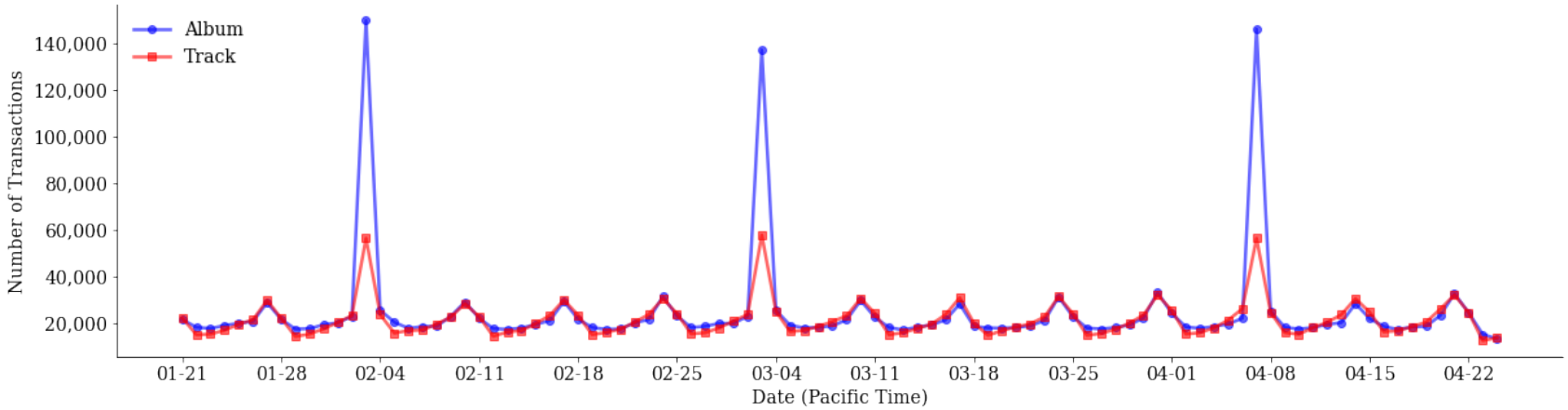
Checking dynamics



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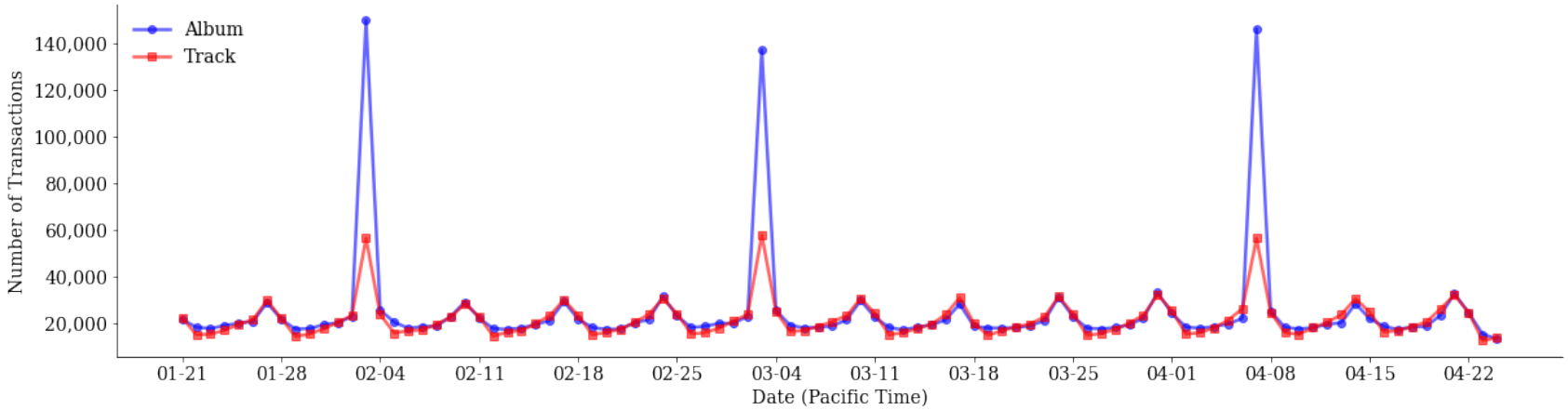

- Do Bandcamp Fridays shift demand?
- Hard to detect dynamics from the figure.
- A finite horizon dynamic model to double check.

A two-stage choice model: First stage

- Consumers make choice among products based on the minimum prices, similar to the general choice model

$$u_{ijt}^{\mathrm{I}} = \boldsymbol{X}_{jt}\boldsymbol{\gamma}_i + \xi_{jt} - \alpha_i \bar{p}_{jt} + \epsilon_{ijt}^{\mathrm{I}}$$

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  - The error term follows the extreme value type I distribution

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$$u_{ijt}^{\mathrm{II}} = \boldsymbol{X}_{jt}\widehat{\boldsymbol{\gamma}}_i + \widehat{\xi}_{jt} - \widehat{\alpha}_i p_{ijt} - \beta_i \max\{r_{ijt} - \delta_t p_{ijt}, 0\} + \epsilon_{ijt}^{\mathrm{II}}$$

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  - $\beta_i \geq 0$ measures consumer i's fairness concern

Choices of payments

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- It can be shown that if $\widehat{\alpha}_i > \beta_i \delta_t$, consumer i chooses to pay the amount that ensures the artist to receive the perceived fair price:

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- Otherwise, consumer i chooses to pay the minimum price

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Identification

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  - Use the variation of δ_t generated by Bandcamp Fridays to identify $f(\beta_i)$

Identification

- How to identify the distribution of consumers' fairness concerns in the market?
  - Use the variation of δ_t generated by Bandcamp Fridays to identify $f(\beta_i)$
- How to identify the distribution of fair prices perceived by fair-minded consumers?
  - Use the variation of $p_{ijt} > \bar{p}_{jt}$ to identify $\mathbb{E}[r_{ijt}|\beta_i > \widehat{\alpha}_i/\delta_t]$

Thank you!

Chen, Y., Koenigsberg, O., and Zhang, Z. J. (2017). Pay-as-you-wish pricing. *Marketing Science*, 36(5):780–791.

Dufwenberg, M. and Kirchsteiger, G. (2004). A theory of sequential reciprocity. *Games and Economic Behavior*, 47(2):268–298.

Fehr, E. and Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *The Quarterly Journal of Economics*, 114(3):817–868.

Gneezy, A., Gneezy, U., Nelson, L. D., and Brown, A. (2010). Shared social responsibility: A field experiment in pay-what-you-want pricing and charitable giving. *Science*, 329(5989):325–327.

Kim, B. C., Park, S. E., and Straub, D. W. (2022). Pay-what-you-want pricing in the digital product marketplace: A feasible alternative to piracy prevention? *Information Systems Research*, 33(3):784–793.

Kreps, D. M., Milgrom, P., Roberts, J., and Wilson, R. (1982). Rational cooperation in the finitely repeated prisoners' dilemma. *Journal of Economic theory*, 27(2):245–252.

Schmidt, K. M., Spann, M., and Zeithammer, R. (2015). Pay what you want as a marketing strategy in monopolistic and competitive markets. *Management Science*, 61(6):1217–1236.